Subject Description Form

Subject Code	EEE532
Subject Title	Artificial Intelligence of Things Technology
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Students are expected to have some basic knowledge on engineering mathematics and electronics.
Objectives	This subject will be offered to cover the knowledge of Artificial Intelligence of Things (AIoT) including sensors, actuators, signal processing, control, Artificial Intelligence and simulation. To support the adoption of AIoT technologies for smart city applications, system design, communication, system model, control theory, AI analytics, Mobile computing, Edge AI and Cloud computing for AIoT system will be taught to make an integrated system for different smart city applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand theory about i) sensors and sensor interface for automated systems. ii) actuators and control for human-computer interfaces and teleoperated systems. iii) different Artificial Intelligence algorithms and their applications in Smart City. iv) system models, signal processing, communication protocol, control theory and AI analytics for their different smart city applications. v) mobile computing, edge AI and cloud computing for AIoT System b. Apply practical skill including coding and system design for AIoT system c. Develop an industrial mindset and understanding in real life applications d. Foster creativity and system design e. Enhance documentation and presentation skill
Subject Synopsis/ Indicative Syllabus	1. Introduction to AIoT and Smart City - Foundational overview of the AIoT (Artificial Intelligence of Things) and its significance in the development of smart cities - The convergence of AI and IoT technologies, the scope of AIoT applications, and the key concepts that define smart city infrastructures and services - Smart city blueprint 2. AIoT System Design and Architecture - Layered architecture, components, and design principles that enable effective deployment of AIoT solutions - System integration, interoperability, and the role of data analytics in enhancing system functionalities 3. IoT Sensors - The types and functions of IoT sensors that gather critical data from the environment - Role of sensors in the AIoT ecosystem, including sensor selection, deployment challenges, and their applications across various domains within smart cities 4. Signal Processing for IoT Sensors - The signal processing methods used to filter and interpret the data collected by IoT sensors

	- The algorithms and techniques employed to enhance data accuracy and reliability 5. IoT Actuators - Functions of actuators as vital components in AIoT systems - Various types of actuators, their operational principles, and their applications in smart environments such as automated traffic signals and smart energy systems 6. Communication of AIoT System - Importance of effective communication protocols for the interoperability of AIoT systems - Various communication technologies, including communication protocol, BLE, WiFi, LoRa, Zigbee, and 5G, that enable robust connectivity and data exchange between devices in a smart city context 7. Control for AIoT System - Control theory for managing AIoT systems - PID control for real-time control to enhance system efficiency and responsiveness within smart cities 8. System Modeling and Simulation for AIoT System - System modeling and simulation techniques that assist in the design and testing of AIoT systems 9. Mobile Computing for AIoT System - Mobile computing to enable access to AIoT applications on the go - Mobile frameworks, application design, and how they integrate with AIoT systems to support mobility and user engagement 10. Edge Computing for AIoT System - Paradigm of edge computing - - How to process data at the edge of the network to reduce latency and improve response times 11. Cloud Computing for AIoT System - Cloud computing in supporting AIoT infrastructures 12. Smart City Applications of AIoT - Various real-world applications of AIoT in smart cities such as smart mobility, smart living, smart environment, smart people, smart government, smart economy
Teaching/Learning Methodology	This subject is designed by 4 levels of learning: 1) Lecture: Focus on theories and knowledge teaching about AIoT technologies including AI, IoT sensor and actuator, communication protocol, system design, control theory, AI algorithms, Mobile computing, edge AI, and cloud computing. 2) Guest Lecture: 3 industrial leaders to share different smart city applications using AIoT 3) Tutorial: 6 tutorial sessions to cover practical skill about ThingsBoard, ESP32 simulator, data collection, system model and communication, data visualization and control. 4) Project-based learning: Project report and Presentation will be conducted by student to learn how to apply the technologies learn from lecture and tutorials for a daily life project. Presentation skill will be further enhanced.

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Teaching/Learning Methodology	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
	a	b	c	d	e
1. Lecture	√		√		
2. Guest Lecture		√	√		
3. Tutorial		√			
4. Project		√		√	√

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<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. Assignments	20	√		√		
2. Project	20		√		√	√
3. Mid-term exam	20	√		√		
4. Final exam	40	√		√		
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Assignments are designed to enable students review the taught materials and take action to practice to learn more in depth of theories.

Project links up with tutorial to provide the guidance of learning practical skill to design and build the AIoT system.

Project Presentation and report provides a chance for student to learn how to share their creativity and present professionally for their idea including speaking and documentation.

Closed book Mid-term exam and final exam are used to assess the learning performance of the students without any assistant by tools such as AI tools.

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<tr><td>Student Study Effort Expected</td><td>

Class contact:	
▪ Lecture	27 Hrs.
▪ Guest lecture	3 Hrs.
▪ Tutorial	9 Hrs.
Other student study effort:	

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<tr><td rowspan="3"></td><td>▪ Lecture</td><td>45 Hrs.</td></tr>
<tr><td>▪ Assignment and Project</td><td>30 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>114 Hrs.</td></tr>
<tr><td>Reading List and References</td><td colspan="2">Reference books:
1. Fadi Al-Turjman, Fahriye Altinay, Zehra Altinay Gazi, “Artificial Intelligence of Things (AIoT)”, Released September 2024, Publisher(s): Morgan Kaufmann, ISBN: 9780443264832
2. Kashif Naseer Qureshi, Thomas Newe, “Artificial Intelligence of Things (AIoT)“, May 2024, Publisher(s): CRC Press, ISBN: 978-1032552996
Reference:
1. The Smart City Blueprint for Hong Kong: https://www.smartcity.gov.hk/
2. Smart City Consortium (SCC): https://smartcity.org.hk/
3. Smart City by ASTRI – Hong Kong Applied Science and Technology Research Institute: https://www.astri.org/technology/smart-city/
4. Smart City Platform with big Data by HKSTP (Hong Kong Science and Technology Park): https://www.hkstp.org/en/our-stories/technology-and-innovation/smart-city-platform/
5. Smart City 3.0 by SCC: https://smartcity.org.hk/en/info-smartcity.php</td></tr>
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